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Computational design of ESIPT-based molecular photoswitches: Electronic structure and quantum transport properties of salicylideneaniline derivatives

Pengcheng Zhang [a,*], Wenjie Shi [a], Songfang Han [a], Zhendong Li [a], Yubing Si [b]

[a] College of Chemistry and Chemical Engineering, Zhoukou Normal University, Zhoukou 466000, People's Republic of China
[b] College of Chemistry, Zhengzhou University, Zhengzhou 450001, People's Republic of China

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ABSTRACT

Excited-state intramolecular proton transfer (ESIPT) offers a unique pathway for designing ultrafast molecular photoswitches with distinct geometric reorganization. In this computational study, we investigated salicylideneaniline (SA) and its thiol-substituted derivatives in the quantum transport systems. The electronic structure and conductance properties of three anchoring configurations were systematically explored. The ESIPT-driven enol-keto tautomerization induces significant geometric planarization and narrows the gap between the highest occupied molecular orbital (HOMO) and the lowest unoccupied molecular orbital (LUMO), while the ESIPT barrier is nearly barrierless (~5 meV), enabling sub-picosecond switching. Transport calculations reveal that destructive quantum interference in the off state yields high on/off ratio (~500 for P-type and ~ 2500 for Z2-type), significantly exceeding those reported for other molecular switch systems. This study establishes ESIPT-based SA derivatives as highly promising candidates for next-generation optoelectronic memory and logic devices.

1. Introduction

Molecular electronics aims to miniaturize electronic components to the single-molecule scale [1,2], with photoswitches serving as critical functional units to precisely modulate electron transport in response to external light stimuli. Traditional photoswitching architectures [3–7], predominantly based on photochromic systems like azobenzenes [8,9] and diarylethenes [10,11], achieve conductance switching via reversible ground-state isomerization. However, these systems are intrinsically plagued by persistent performance trade-offs. For instance, azobenzenes suffer from thermal instability of the *cis*- isomer and relatively low on/off ratios, while diarylethene derivatives, despite excellent thermal bistability, typically exhibit strong molecule-electrode electronic coupling. This coupling efficiently quenches the excited state of the open-ring form when directly tethered to the electrodes. Moreover, both systems undergo large structural changes during isomerization, which can lead to junction rupture in single-molecule devices, and they are prone to photodegradation after limited switching cycles. Consequently, simultaneously achieving high switching ratio, rapid photo-response speed, and robust operational stability remains a challenge in molecular junction design.

ESIPT [12–15] has emerged as a highly promising photochemical paradigm for conducting high-performance molecular devices. This intramolecular proton migration not only drives subtle geometric planarization, but more importantly, induces a profound redistribution of molecular electron density and spatial localization/delocalization of frontier orbitals. Salicylideneaniline and its derivatives [16–18] are prototypical ESIPT systems that undergo ultrafast tautomerization upon excitation, converting from the enol to the keto form. This process is particularly attractive for molecular electronics because it combines rapid photo-response with distinct geometric changes: a shift from sp^2 hybridized imine bridge in the enol form to sp^3 hybridized amine bridge in the keto form. This structural disparity suggests a potential for significant modulation of π-conjugation and, consequently, molecular conductance [19].

Recent breakthroughs have begun to harness ESIPT in single-molecule junctions [5,20]. Notably, 2-(2-hydroxyphenyl)pyridine derivatives with para- and meta-thiomethyl anchoring groups have demonstrated bidirectional photoconductance modulation [21]: one scaffold exhibited a record of ~120-fold conductance decreases under

* Corresponding author.
E-mail address: pczhang@zknu.edu.cn (P. Zhang).

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365 nm irradiation due to frontier orbital localization dominating over bandgap narrowing, while the other showed an ~1.78-fold enhancement driven by a transition from destructive to constructive quantum interference (QI) [22,23]. However, translating the ESIPT mechanism into functional molecular junctions remains a challenge. The electronic coupling between the molecule and the metal electrodes, as well as the specific anchoring geometry, can drastically alter the intrinsic electronic properties of the switch. Previous studies [18,24] have primarily focused on the spectroscopic properties of SA derivatives in solution or crystalline phases, leaving a critical gap in understanding how the ESIPT-driven conductance switching occurs in thiol-anchored systems confined between gold electrodes.

To the best of our knowledge, no study to date has systematically investigated how the thiol anchoring configurations affect the QI characteristics and light-controlled switching performance of ESIPT-based single-molecule junctions. Therefore, we focus on the structure-activity relationship by elucidating the electronic structure changes induced by ESIPT and quantify the quantum transport properties across different anchoring configurations (P, Z1 and Z2 types). In this study, density functional theory (DFT) combined with Non-Equilibrium Green's Function (NEGF) techniques are employed to systematically investigate the electronic properties of thiol-substituted SA derivatives in the molecular photoswitches.

The results uncover strong configuration-dependent switching behavior, with junctions exhibiting exceptionally high on/off ratios, thereby establishing SA derivatives as promising candidates for high-performance optoelectronic switches. Additionally, the technical advantages of this research directly address the core bottleneck in the industrialization of single-molecule electronic devices, namely the difficulty of simultaneously achieving high switching ratio, ultra-fast response, and long-term structural stability. High-density single-molecule storage urgently requires both high signal-to-noise ratio (depending on high switching ratio) and low write energy consumption (depending on ultra-fast response), while ultra-low-power molecular logic circuits highly rely on the photonic switch not undergoing covalent bond breakage or conformational distortion during repeated cyclic operations. The ESIPT molecular junction designed in this work can simultaneously meet these multiple stringent requirements, thereby significantly enhancing the application value.

2. Computational methods

All electronic structure calculations for clusters were performed within ORCA 5.0.4 package [25], using the range separated hybrid exchange-correlation functional ωB97X-D3 [26,27] and the def2-TZVP basis set. Solvent effects were mimicked via the conductor-like polarizable continuum model (CPCM) [28] with cyclohexane as the solvent. Ground-state geometry optimizations were carried out for both enol and keto forms of SA and its thiol substituted derivatives. Vibrational frequency analyses were conducted to confirm that all optimized structures correspond to local minima, as indicated by the absence of imaginary frequencies. The Gibbs free energy profiles for the ESIPT process were calculated based on the harmonic oscillator approximation at 298.15 K and 1 atm. Specifically, G was derived from frequency calculations and expressed as: $G = E_{elec} + ZPE + E_{vib} + E_{rot} + E_{trans} + k_BT - TS$, where E_{elec} is the electronic energy, ZPE represents the zero-point energy, k_B is Boltzmann constant, T represents temperature, and S represents entropy. Excited-state calculations were performed using time-dependent DFT (TD-DFT) at the same theoretical level. Vertical excitation energies were obtained to simulate absorption spectra, and potential energy curves along the proton transfer (PT) coordinate were constructed by varying the O—H distances from 0.95 to 1.80 Å. The excitation analyses were performed using the Multiwfn version 3.8(dev) toolkit [29], while molecular structures and frontier orbitals were visualized with VESTA [30].

The optimized molecule is sandwiched between Au electrodes to form the single-molecule junction. The thiol anchoring groups were deprotonated to form thiolate bonds with gold adatoms at the electrode surfaces. The molecular junctions were then further optimized, and the electronic structure was calculated using the program SIESTA 5.2.2 [31]. The computational model for the Au(001)-molecule junction follows the protocol established in recent studies on similar single-molecule junctions [6], which has been validated to reliably reproduce the anchoring geometry and electronic coupling of thiol-gold interfaces. The molecular junction was modeled as a two-probe system consisting of a central scattering region (the SA molecule) connected to two semi-infinite gold electrodes. Specifically, the electrodes were modeled as Au(001) surface supercells (4 × 4) with a side length of 11.5 Å. During optimization, the last 5 layers of Au atoms were fixed, while part of the surface gold atoms (including two complete layers of Au atoms extended from the bulk and five Au atoms forming a square pyramidal tip) in the scattering region and the molecular atoms were allowed to relax. The initial Au—S thiolate bond length was set to about 2.4 Å based on empirical values. The PBE functional with TZP basis sets was used for transport calculations. The mesh cutoff energy is set to 500 Ry. A 3 × 3 × 100 k-point grid was employed for electrode calculations, while a 3 × 3 × 1 grid was used for the scattering region. The convergence criterion for the density matrix is established at 1.0×10^{-4}, and the force convergence criterion is set to be 0.05 eV/Å. All the above fundamental parameter settings have been verified through convergence tests, as detailed in Supplementary Information (SI), Section S1, Table S1.

Based on the optimized structure and electronic properties, the transport characteristics of the molecular junctions are investigated. The NEGF method combined with DFT, as implemented in TranSIESTA and TBtrans [32,33], is utilized for this purpose. The same exchange-correlation functional and basis set parameters used in the structure optimization are applied here. The chemical potentials and the electron transmission coefficient are key parameters in determining the current. The current through the molecular junction is calculated using the Landauer-Büttiker formula [6,34]:

$$I(V) = \frac{2e^2}{h}\int_{\mu_L}^{\mu_R} T(E,V)[f(E-\mu_L) - f(E-\mu_R)]dE \quad (1)$$

In this equation, e is the charge of the electron, h is Planck's constant, $T(E, V)$ is the electron transmission coefficient, and $f(E\text{-}\mu_{L/R})$ is the Fermi-Dirac distribution function of the electrons in the left/right electrodes with chemical potentials $\mu_{L/R}$, respectively. Current-voltage (I-V) characteristics were computed by integrating the transmission function $T(E)$ over the bias window. The bias voltage was scanned from 0.1 to 1.0 V in steps of 0.1 V, with $T(E)$ sampled at 0.01 eV intervals across an energy range of ±3 eV around the Fermi level. All calculations were performed at zero temperature unless otherwise noted.

The distinct computational setups employed here reflect the differing physical demands of each simulation stage. The ωB97X-D3/def2-TZVP level in ORCA is chosen to maximize accuracy for excited-state energetics and dispersion-bound geometries in isolated molecules. Conversely, the PBE/TZP setup in SIESTA is adopted for its established reliability, numerical stability, and efficiency within the NEGF framework for periodic metal-molecule-metal junctions. This dual-level approach ensures both spectroscopic precision and transport robustness.

3. Results and discussion

3.1. Electronic structure and ESIPT mechanism of SA

To establish a foundation for molecular junction, the geometric and electronic structures of the pristine SA molecule should be elucidated first. As depicted in Fig. 1 and Table 1, the optimized ground-state geometry reveals a significant structural transformation from enol to keto form. While the ∠C2C3-NC4 dihedral angle remains planar in both

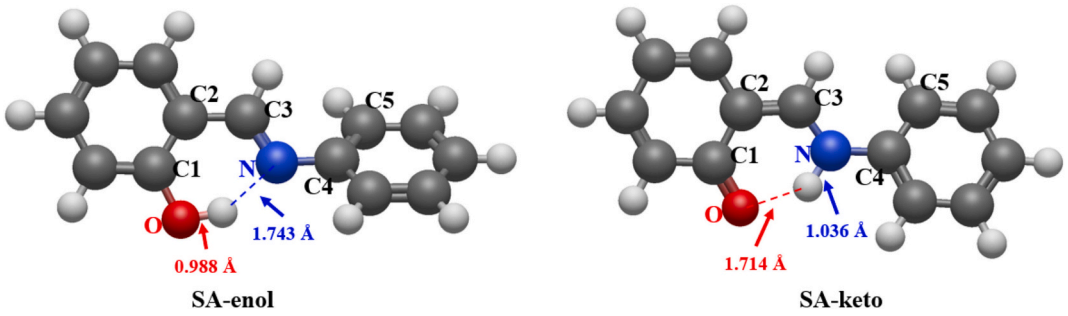


Fig. 1. The optimized structures of SA at ωB97x-D3/def2-TZVP level. The C, N, O, H atoms are represented by grey, blue, red and white, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 1
The geometric parameters in two forms of SA.

SA	Enol	Keto
Bond length C3—N (Å)	1.276	1.316
Bond length C1—O (Å)	1.336	1.256
Dihedral ∠C2C3-NC4 (°)	177.6	179.2
Dihedral ∠C3N-C4C5 (°)	43.6	18.4

tautomers, a dramatic change occurs in the ∠C3N-C4C5 dihedral angle. In the enol form, the sp^2 hybridized nitrogen results in a twisted structure with a dihedral angle of 43.6°. After PT to the keto form, the nitrogen atom transitions to sp^3 hybridization, which leads to a significant reduction in steric hindrance. Consequently, the keto tautomer adopts a much more planar configuration, with the dihedral angle decreasing to 18.4°. This enhanced planarity is crucial as it facilitates better π-electron delocalization across the molecular backbone in the keto form.

The alteration in geometry is also reflected in the frontier molecular orbitals. The molecular orbitals energy levels and frontier orbitals of SA have been plotted in Fig. 2. It shows that the keto tautomer exhibits a substantially narrowed HOMO-LUMO energy gap (7.28 eV) compared to the enol form (8.42 eV). This reduction in the energy gap is consistent with the simulated absorption spectra (Fig. S1: absorption spectra of pristine SA enol and keto forms), where the S_1 excitation of the keto form is red-shifted to 3.49 eV, in contrast to 4.32 eV for the enol form. This indicates that despite the sp^3 hybridization of the bridging nitrogen, the keto tautomer possesses a more effective conjugation length, likely stabilized by a quinoid-like resonance structure.

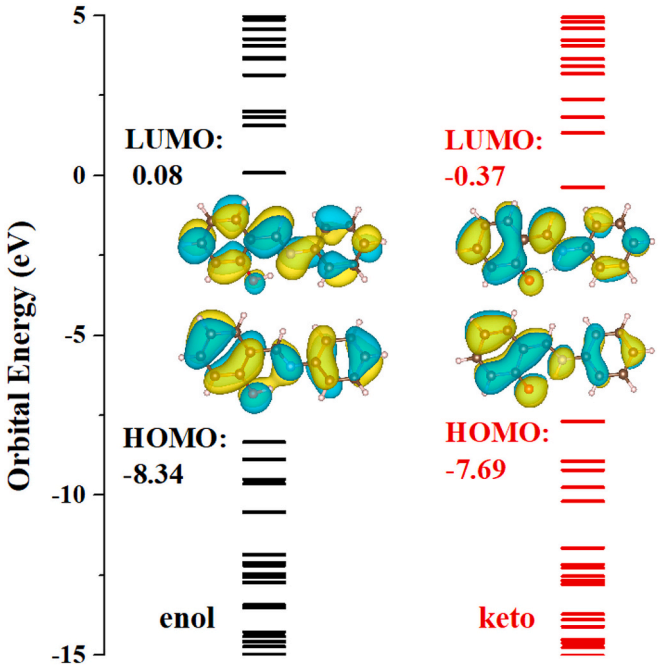


Fig. 2. Molecular orbitals energy levels and frontier orbitals (HOMO and LUMO) of the enol and keto forms (Isovalue = 0.02).

In the ground state, the enol form is thermodynamically favored, and the PT process passes through a transition state (TS) with a barrier of ΔG = 0.192 eV. In contrast, the energy barrier for ESIPT via the excited state S_1 is typically lower [15]. The S_1 ESIPT potential energy surface was constructed using constrained geometry optimization (default RFO algorithm in ORCA). During the scan, the O—H bond length was varied with a uniform step size of 0.05 Å, while the dihedral angle ∠C2C3-NC4 was kept fixed and all other degrees of freedom were fully relaxed at each constrained point. The resulting energy profile is plotted in SI, Section S3, Fig. S3 (potential energy curves for PT on S_0 and S_1 states). The molecule readily undergoes a nearly barrierless pathway in S_1, with the calculated electronic energy barrier being merely 5 meV. This ultrafast, almost adiabatic proton transfer process confirms the potential for sub-picosecond switching speeds in this system.

Furthermore, the above calculations suggest that the experimentally observed dual fluorescence likely originates from the distinct environments. While the keto form exhibits strong fluorescence (emission in 2.92 eV), the enol form in isolation tends to relax via a twisted intramolecular charge transfer (TICT) state [35] (1.15 eV) with a neglectable oscillator strength. It is hypothesized that the dual fluorescence observed in experiments is predominantly a solid-state or aggregated phenomenon [24,36], where steric constraints imposed by the crystal lattice restrict the torsional motion (∠C2C3-NC4 dihedral rotation), thereby locking the enol form and allowing its emission to be observed. The overall energy diagram is listed in Fig. 3.

3.2. Molecular design of thiol-substituted derivatives

To realize SA-based molecular junctions, the SA molecule must be anchored between gold electrodes. Since the hydrogen atom of the thiol group dissociates upon binding to the gold electrode, three types of -S-Au substituted SA molecules (SA-S) were directly constructed: one para-substituted (P) and two zigzag-substituted (Z1, Z2) configurations. The optimized structures are shown in Fig. 4. The structure parameters such as bond lengths and dihedral angles are listed in Table S2. These configurations were strategically chosen to modulate the electron transport pathway across the molecular backbone. For the P-type substitution, the current pathway traverses both benzene rings, maintaining a linear conjugation. For Z1 and Z2 substitution, the conductance pathways deviate from the direct benzene-benzene connection. Specifically, in the Z1 configuration, the pathway involves the O-H-N hydrogen bond, which is critical in ESIPT. Whereas in Z2, the pathway bypasses the hydrogen bond structure.

The introduction of thiol anchor groups and their binding to gold atoms significantly perturbs the electronic structure of the pristine SA molecule. As shown in Fig. 5, the hybridization between the sulfur lone

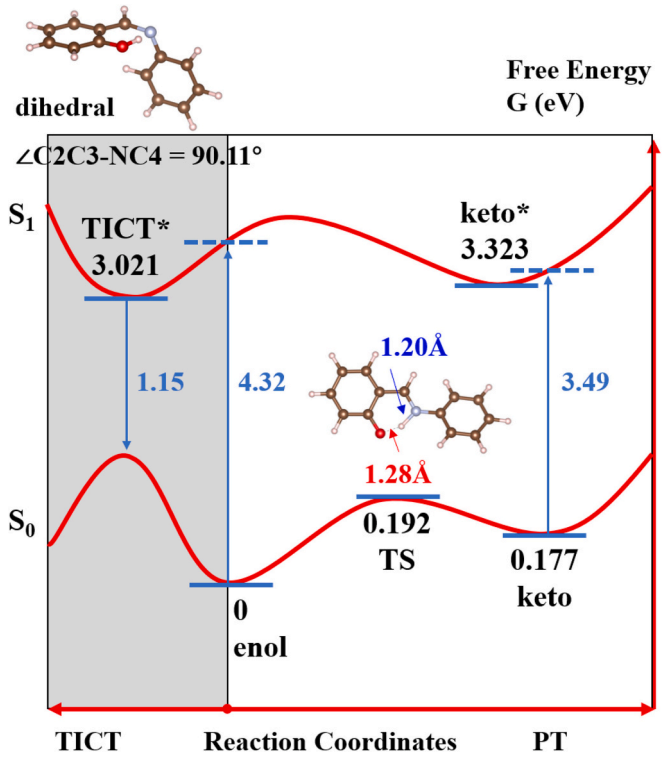


Fig. 3. Energy diagram on ground state and excited state (S_1) of transitions between two forms.

pairs and gold orbitals introduces new electronic states within the original HOMO-LUMO gap, effectively reducing the energy gap from approximately 7.5 eV in the isolated SA to around 6.5 eV in the SA-S derivatives. Despite this reduction, the trend observed in the parent system persists: the keto tautomers consistently exhibit narrower HOMO-LUMO gaps compared to their enol counterparts. This indicates that the keto forms maintain superior π-conjugation even after functionalization.

To validate the photophysical integrity of these derivatives, the absorption spectra of all SA-S derivatives were calculated and shown in Fig. S4 (enol form) and S6 (keto form). The excited-state charge difference density (CDD) of selected state has also been plotted (Fig. S5 and S7: CDD distributions of S_9 for enol forms and S_5 for keto forms). The absorption profiles of the enol and keto forms exhibit distinct features corresponding to their electronic transitions. For the enol forms, the low-lying excited states are primarily localized on the anchoring S and Au atoms. Specifically, the original S_1 excitation of pristine SA (with CDD localized on the salicylaldehyde ring and C═N bond) shifts to higher energy states in SA-S derivatives. It corresponds to the S_9 state in the enol form, and S_5 state in the keto form. This shift in the dominant excited state (from S_1 in SA to S_5/S_9 in SA-S) is attributed to the perturbation induced by the heavy atoms (S, Au). Nevertheless, the distinct electronic reorganization in the excited state confirms that the photo-triggered proton transfer mechanism remains operative.

The ground-state PT energetics to evaluate the thermal stability of the switching states within the electrode environment were further investigated. The potential energy profile was constructed via constrained geometry optimization, where only the two anchoring Au atoms at the electrode ends were frozen while all other degrees of freedom were fully relaxed at each scan point along the O—H coordinate (step size 0.05 Å). As shown in Fig. 6, the PT barriers for the thiol-substituted systems (0.245–0.277 eV) are slightly lower than that of unsubstituted SA (0.29 eV). More importantly, the stability of the keto tautomer is enhanced upon anchoring: the relative energies of keto configurations (0.132, 0.159 and 0.168 eV) are lower, compared to 0.177 eV in the

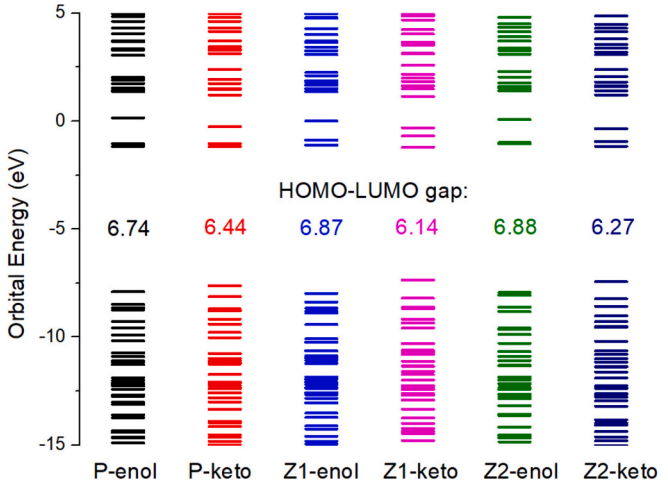


Fig. 5. Molecular orbitals energy levels and HOMO-LUMO gaps of SA-S derivatives.

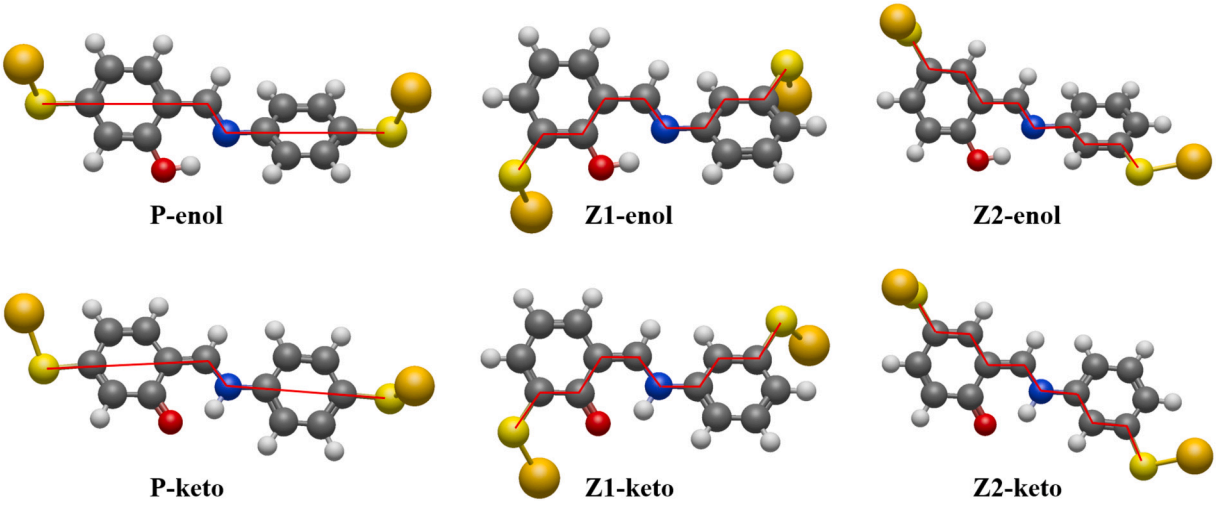


Fig. 4. The optimized structures of SA-S derivatives. The conductance paths are lined with red. S and Au atoms are represented by light and dark yellow, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

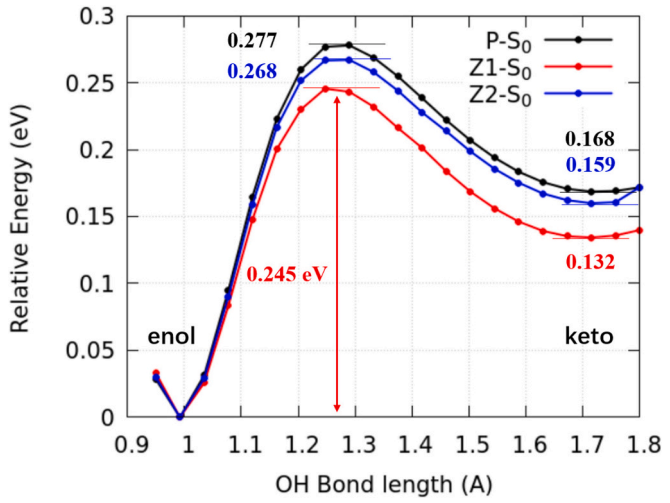


Fig. 6. The scanned potential energy curves of PT process for ground-state SA-S derivatives.

isolated molecule. This thermodynamic stabilization suggests that both on and off states of the SA-S molecular junctions are robust.

To bridge the calculated energy barriers with assessable device-level parameters, we further estimated the thermal isomerization kinetics using the Arrhenius relationship (see SI, Section S6, Table S3: Arrhenius kinetic parameters for proton transfer). The analysis reveals that the reverse rate constant (k_R) for the keto-to-enol relaxation exceeds 10^{11} s^{-1} at 300 K, ensuring that the high-conductance keto state thermally relaxes back to the stable enol form on a sub-nanosecond timescale once illumination ceases. This rapid ground-state recovery prevents thermal accumulation of the conducting state, guaranteeing robust dark-state stability. Conversely, the nearly barrierless ESIPT in the excited state (< 5 meV) corresponds to a forward switching rate exceeding 10^{12} s^{-1}, enabling sub-picosecond optical turn-on. These kinetic metrics confirm that SA-based junctions simultaneously possess ultrafast photo-response and excellent thermal bistability.

3.3. *Quantum transport properties*

The focus is now shifted to the quantum transport properties of the SA-based photoswitches. The structural configurations of the SA-S molecular junctions, where the molecule is sandwiched between gold electrodes, are illustrated in Fig. 7 (P-type) and Fig. S8, S9 (Z1- and Z2-type). The electrode tip model where a single gold atom at the tip binds to the sulfur atom were constructed. After structural optimization, the device region shows minimal deviation from the cluster structure. The keto isomer still exhibits a smaller dihedral angle compared to the enol form.

Fig. 8 presents the zero-bias transmission spectra for the three anchoring configurations, revealing markedly distinct electronic behaviors. In contrast to their tautomeric counterparts, the P-enol, Z1-keto, and Z2-enol isomers all exhibit smooth, featureless transmission curves with severely suppressed conductance near the Fermi level. This pronounced transmission suppression is a hallmark of destructive QI

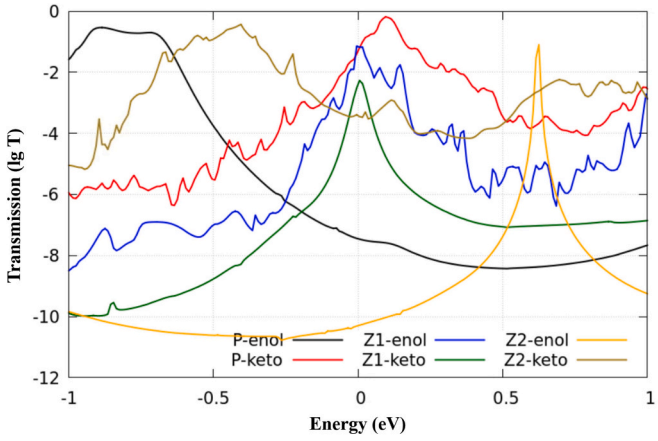


Fig. 8. The transmission spectra of P, Z1, Z2 type of molecular junctions both in enol and keto forms (presented in logarithmic scale).

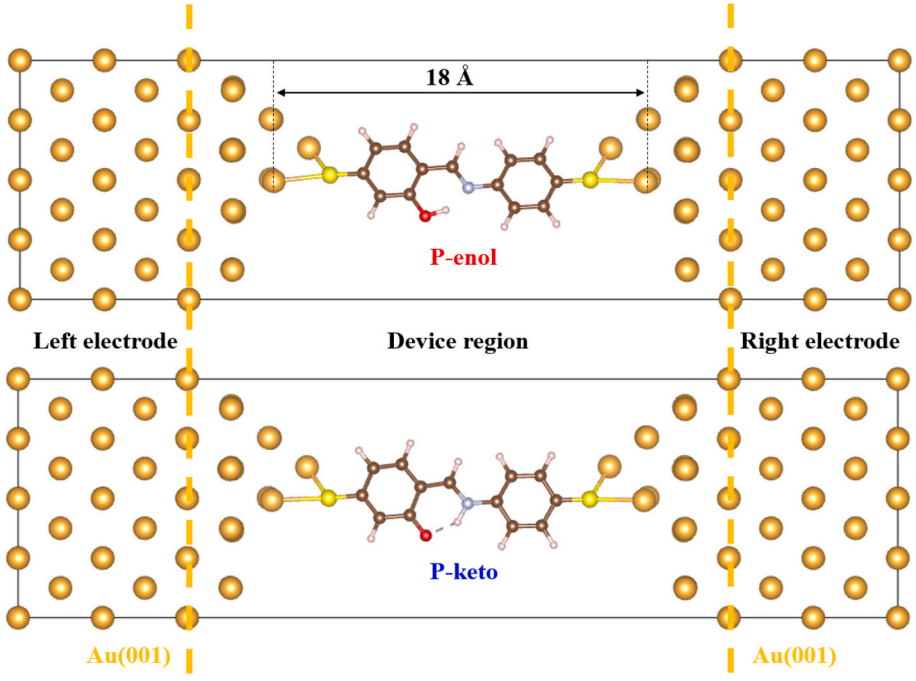


Fig. 7. The molecular junction structures of P substitute SA in enol and keto forms. The left and right electrodes extend indefinitely, and the device region includes the molecular and the buffer layer.

[23], where electron wave functions propagating through different molecular pathways interfere destructively, effectively canceling the net transmission probability. At zero bias, both P-enol and Z2-enol are situated within a distinct low-conductance window, establishing a well-defined off state. This QI-induced transmission dip serves as the physical origin of the off state and, critically, provides the essential precondition for achieving an extraordinarily high on/off switching ratio. Furthermore, the ultra-high switching ratio in Z2-enol is fundamentally rooted in its geometric asymmetry. Unlike the highly symmetric P-enol configuration, the zigzag anchoring geometry in Z2-enol breaks the spatial symmetry of the molecular junction (see SI, Section S8, Fig. S10: spatial distributions of VBM and CBM in periodic P-enol and Z2-enol junctions). This structural asymmetry leads to an unbalanced electronic coupling between the molecular backbone and the two electrodes, causing the frontier states to exhibit pronounced spatial localization and asymmetric spatial distribution. Such geometric distortion severely suppresses the effective orbital overlap along the transport path, thereby generating a deeper and broader destructive interference valley near the Fermi level.

The calculated I-V characteristics (Fig. 9a) show that the current increases significantly for P-keto, Z1-enol, and Z2-keto with rising bias voltage. For the P and Z2 configurations, the conductance of the keto form is substantially higher than that of the corresponding enol form across the entire bias range. In contrast, for the Z1 configuration, the current for keto increases slowly and remains comparatively low at 1 V. Consequently, the enol form represents the high-conductance on state, while the keto form constitutes the low-conductance off state. The inverted switching behavior of the Z1 configuration may be tentatively attributed to its unique anchoring geometry. The ortho-positioned thiol anchor likely directs the electron transport pathway through the O—H···N hydrogen-bond region. Upon ESIPT, the proton transfer induces geometric distortion in this region, which could alter the local orbital phase and electronic coupling strength, switching constructive interference to destructive interference. While further quantitative analysis is needed to fully validate this mechanism, it highlights the possible role of anchoring position in modulating hydrogen-bond-mediated quantum interference pathways.

The on/off ratio, defined as $R = I_{high}/I_{low}$, serves as a critical metric for evaluating switching performance. As shown in Fig. 9b, three types of junctions show distinctive characters. P-type achieves a high switching ratio of approximately 500 at low bias. However, this ratio gradually decreases as the voltage increases to 1 V. Z1-type exhibits the smallest switching ratio, suggesting minimal conductance difference between the two tautomers in this specific geometric configuration. The ratio of Z2-type shows a peak at 0.7 V, achieving the highest switching ratio up to ~2500. This superior performance is attributed to the combination of a highly conductive keto state and an extremely suppressed transmission in the Z2-enol state. For comparison, the highest switching ratio reported for fulgimide-based molecular junctions is 853 at −0.2 V [6], and the record photoconductance modulation for ESIPT-based single-molecule junctions is approximately 120-fold [21]. For azobenzene-based molecular junctions, theoretical calculations indicate switching ratios typically below the order of 10^1 level [8]. The on/off ratio of ~2500 achieved by the Z2-type junction in this work significantly exceeds these reported values, representing a substantial advancement in the switching performance of molecular photoswitches.

To facilitate direct comparison across the three anchoring configurations, key electronic and transport parameters are summarized in Table 2. The cluster-level HOMO-LUMO gaps and PT barriers characterize the intrinsic molecular properties, while the currents at bias voltage of 1 V and peak on/off ratios reflect the device-level switching performance within the periodic junction. Notably, this table highlights the configuration-dependent switching polarity: P and Z2 operate in a keto-on/enol-off mode with high on/off ratios (~500 and ~ 2500, respectively), whereas Z1 exhibits reversed enol-on/keto-off behavior with markedly lower ratio. This unified presentation underscores the critical role of anchoring geometry in modulating ESIPT-driven conductance switching.

4. Conclusions

In this work, we have systematically investigated the electronic structure and quantum transport properties of SA derivatives using DFT combined with the NEGF technique. Our calculations reveal a distinct ESIPT-driven conductance switching mechanism, which is highly dependent on the molecular anchoring geometry. To maintain a clear distinction between rigorously established theoretical findings and predictions for future experimental validation, the conclusions are presented in the following two complementary layers:

Table 2
Summary of key electronic transport and structural parameters for P/Z1/Z2 configurations (values are given as enol/keto).

Parameter	P	Z1	Z2
HOMO-LUMO gap (eV)[a]	6.74 / 6.44	6.87 / 6.14	6.88 / 6.27
Ground-state PT barrier (eV)[a]	0.277	0.268	0.245
Current at 1 V (nA)[b]	~100 / ~4400	~3000 / ~300	~3 / ~3500
Peak on/off ratio[b]	~500 (0.1 V)	~20 (0.3 V)	~2500 (0.7 V)

[a] From cluster calculations (ωB97X-D3/def2-TZVP), [b] From NEGF-DFT (PBE/TZP).

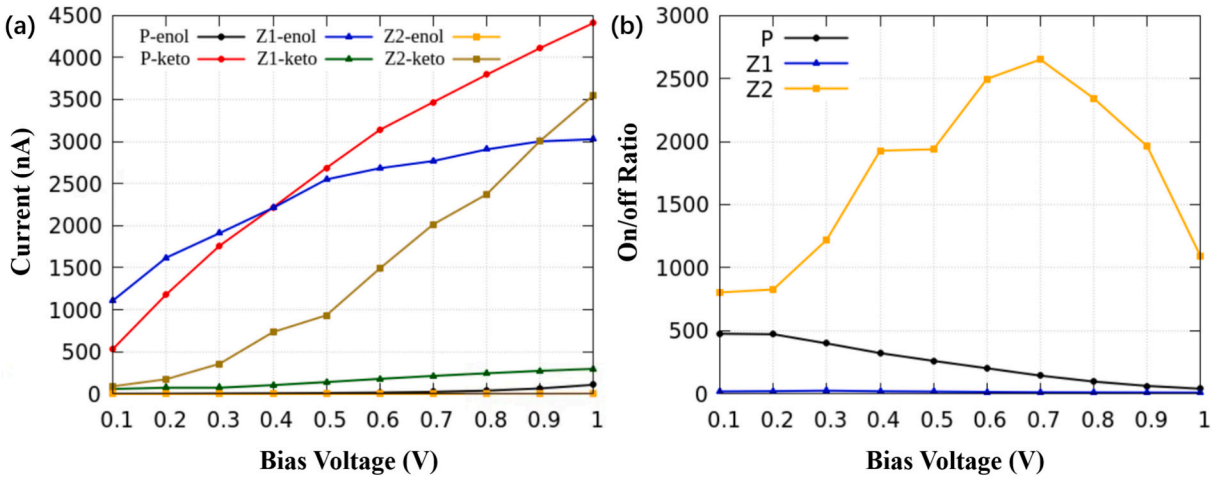


Fig. 9. (a) Current-voltage (I-V) curves and (b) switch ratio as a function of bias voltage (0–1 V) of three types of molecular junctions (P, Z1 and Z2).

(1) The calculations establish that the ESIPT process from enol to keto tautomer induces a pronounced geometric planarization, decreasing the key dihedral angle from 43.6° to 18.4°. This enhances π-orbital delocalization and narrows the HOMO-LUMO gap of the isolated molecule by approximately 1.1 eV. Upon introducing thiol anchoring groups, this relative gap narrowing persists, while the thermodynamic stability of both tautomers is enhanced. The transport calculations further demonstrate that destructive QI is the primary mechanism governing the off-state conductance. Specifically, in the Z2 zigzag configuration, the asymmetric anchoring geometry induces a phase mismatch in the electron transport pathways, resulting in a deep and broad transmission suppression dip near the Fermi level. This interference effect yields a theoretical peak switching ratio of up to ~2500 between the high-conductance keto state and the low-conductance enol state. In contrast, the P-type junction exhibits moderate maximum ratio of ~500, while the Z1 configuration shows an inverted switching polarity with significantly lower ratio. It is demonstrated that the anchoring geometry governs not only the switching efficiency but also the on/off polarity.

(2) The structure-activity relationships established here provide clear, testable predictions for experimental realization. The Z2-type thiol-substituted SA derivative, with its asymmetric anchoring points, is predicted to exhibit an exceptionally high photoswitching ratio when integrated into single-molecule devices. We suggest that this derivative could potentially be synthesized via established substitution protocols and subsequently characterized using scanning tunneling microscope break-junction (STM-BJ) or mechanically controllable break-junction (MCBJ) techniques under light irradiation.

While our conclusions are robust within the theoretical framework, several inherent simplifications should be acknowledged when extrapolating these results to realistic device conditions. First, the electronic structure calculations were performed for isolated molecules using implicit solvent models, which do not capture the dynamic hydrogen-bond rearrangements or polar solvation effects that may modulate the ESIPT barrier and excited-state relaxation pathways in solution-processed devices. Second, the electrode surfaces were modeled as ideal Au(001) facets with a pyramidal tip geometry, neglecting the presence of atomic steps, vacancies, or chemisorbed impurities commonly found in experimental break-junction setups. Such defects could introduce additional scattering centers and alter the effective electronic coupling at the molecule-electrode interface. Third, the transport calculations were performed at zero temperature, disregarding thermal broadening of the Fermi distribution and phonon-assisted inelastic tunneling, which may affect the on/off ratio and switching stability under ambient operating conditions.

CRediT authorship contribution statement

Pengcheng Zhang: Writing – original draft. **Wenjie Shi:** Investigation. **Songfang Han:** Investigation. **Zhendong Li:** Investigation. **Yubing Si:** Writing – review & editing.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.comptc.2026.116023.

Data availability

Data will be made available on request.

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